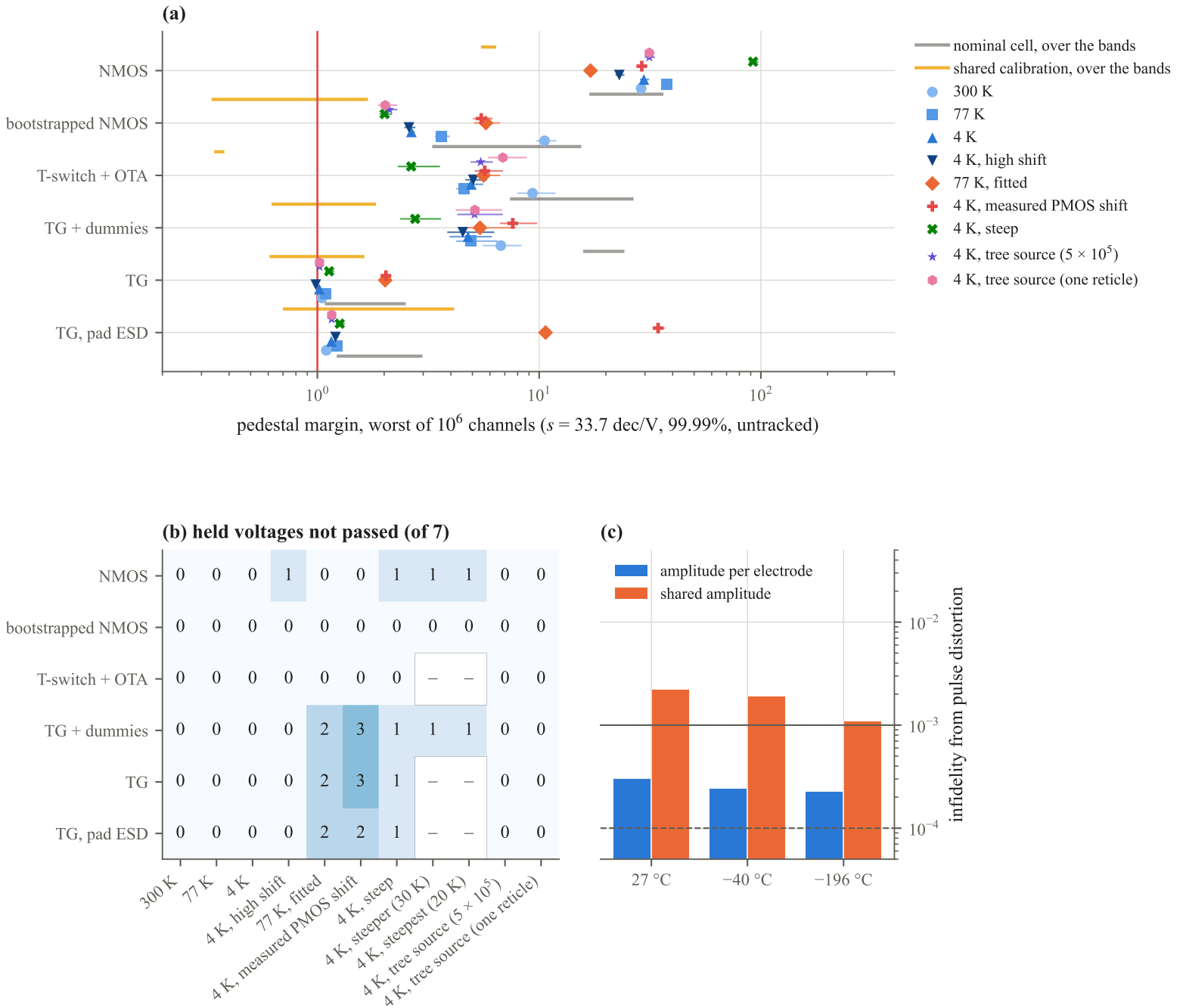
(a)
NMOS
bootstrapped NMOS
T-switch + OTA
TG + dummies
TG
TG, pad ESD
10^0
10^1
10^2
pedestal margin, worst of 10^6 channels (s = 33.7 dec/V, 99.99%, untracked)
nominal cell, over the bands
shared calibration, over the bands
300 K
77 K
4 K
4 K, high shift
77 K, fitted
4 K, measured PMOS shift
4 K, steep
4 K, tree source (5×10^5)
4 K, tree source (one reticle)
(b) held voltages not passed (of 7)
NMOS 0 0 0 1 0 0 1 1 1 0 0
bootstrapped NMOS 0 0 0 0 0 0 0 0 0 0 0
T-switch + OTA 0 0 0 0 0 0 0 – – 0 0
TG + dummies 0 0 0 0 2 3 1 1 1 0 0
TG 0 0 0 0 2 3 1 – – 0 0
TG, pad ESD 0 0 0 0 2 2 1 – – 0 0
300 K
77 K
4 K
4 K, high shift
77 K, fitted
4 K, measured PMOS shift
4 K, steep
4 K, steeper (30 K)
4 K, steepest (20 K)
4 K, tree source (5×10^5)
4 K, tree source (one reticle)
(c)
amplitude per electrode
shared amplitude
10^{-2}
10^{-3}
10^{-4}
infidelity from pulse distortion
27 °C
−40 °C
−196 °C